



HK IMO Training 2023-2024
Problem Set 2
For 7 October 2023



Problem 5. There are n cards in a single pile. In each move, we can choose a pile with at least 3 cards, remove 1 card from this pile, and divide the remaining cards in this pile into two nonempty piles (not necessarily with the same number of cards). Find all positive integers n such that it is possible to make all piles have exactly 3 cards after some moves.

Problem 6. Let $n \geq 2$ be an integer, and let x and y be real numbers. Prove that $x - y$ is an integer divisible by n if and only if

$$\lfloor kx \rfloor \equiv \lfloor ky \rfloor \pmod{n}$$

for every positive integer k .

Problem 7. Let a, b, c be real numbers satisfying $0 < a \leq b \leq c$. Prove that

$$\frac{a+b+c}{3} - \frac{3}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}} \geq \frac{(c-a)^2}{6c}.$$

Problem 8. Let I be the incentre of a scalene $\triangle ABC$. The incircle of $\triangle ABC$ touches BC, CA, AB at D, E, F respectively. Let M and N be the midpoints of DE and DF respectively. The line MN intersects the lines BC and AD at P and Q respectively. Suppose AP meets the circumcircle of $\triangle ABC$ again at S . Let T be the foot of perpendicular from I to AP . Prove that SD and TQ are parallel.